

PRELIM RULES

This section summarizes and distills the department's official policy on the preliminary examination. The next section on this page contains advice from past graduate student's on tackling the prelim process. Please check the 8/8/2023 Student Handbook [https://docs.google.com/document/d/1AXXY5OXicQGNli7OKJoCD5HYGwQ8yNsiHXxJ_f6EVEo/edit?usp=sharing] for official policy!

Overview

The Prelim is an oral examination conducted by three members of the Department appropriate for the fields in which the student will be examined. It consists of three topics. Typically, each topic is allotted ~40 minutes of questions for a total of 2-3 hours (given breaks, discussion time, etc). Students can choose the committee and order in which the exams are administered. The Exam may be taken at any time the student chooses, subject to the regulations listed below. Per Grad Div policy, the Prelim rules are decided by the Astronomy faculty, whereas rules of the Qual (see below) are set by Grad Div.

The primary purpose of the Prelim is for a student to master the fundamentals of a broad slate of topics in astrophysics. A student should speak comfortably, knowledgeably, and quantitatively about the topics tested and demonstrate that their level of understanding is sufficient to understand, and to place in context, a typical department-wide colloquium in astrophysics. Another purpose of the exam is to develop the skill of thinking on one's feet: to perform rough but quantitative estimates in real time, on the board if necessary, and to reason using physics and mathematics.

Timeline

Students must attempt the Prelim by the end of their 2nd summer at Berkeley. The timeline may be slightly different for transfer students and should be discussed with the HGA. The process for scheduling the Prelim is:

1. Fill out the prelim Google Form [<https://forms.gle/29kuKcsWox3cxp2x9>], and get in contact with Yasasha to fill out the paperwork to schedule the prelim.
2. Arrange a meeting with the HGA to discuss Prelim plans.
3. Modify form as necessary.
4. HGA approves the Prelim Application.

The choice of topics and Prelim Committee members must be approved by the HGA at least 3 weeks in advance of the proposed date of the exam. It is recommended that students begin planning for their Prelim (e.g., asking committee members to serve) 1 semester in advance of their planned exam date.

Topics

Students select three topics, generally from their core courses [https://badgrads.berkeley.edu/doku.php?id=academic_rules:core], though they may choose to be examined in another topic at a comparable level of difficulty, subject to HGA approval and the agreement of a senate faculty member to serve as an examiner. Occasionally, the topics of a special topics class (AY 250) may be used as an exam topic. Some non-standard topics have included:

- Stellar Atmospheres
- Star Formation
- Planetary Dynamics
- The Sun
- Stellar Populations

Radiative Processes (C207) and Astrophysical Fluid Dynamics (C202) cannot both be used as prelim topics, on the grounds that a student is not tested on enough phenomenology. Other combinations (e.g., Star Formation and Interstellar Medium) may also be disallowed to ensure that the students develop breadth. The eligibility of a Special Topics course taught as Astro 250 will be reviewed by the faculty on a case-by-case basis. The instructor of the Astro 250 in question should announce at the beginning of the semester whether the course is eligible and what, if any, prelim topic combinations involving the course are prohibited. Courses taken outside of Astronomy are generally not eligible as Prelim topics.

The student will select the order in which the topics are examined.

It is possible for students to take an upper level undergraduate version of a class and use that for the prelim. For example, taking undergrad stars could serve as the basis for a prelim. This is an exception to normal practice and should only be considered in very unusual circumstances. This exception must be approved by the HGA well in advance of taking the prelim and related classes.

Committee

Prelim committee members are usually drawn from all instructors of the core Astronomy graduate courses. They are usually, but not always, members of the Academic Senate. A student's prelim committee member need not have been the student's instructor for a core course. The student should ask one member to serve as chair. Faculty will make good faith efforts to accommodate requests to administer Prelim exams, but they may not be able to do so (e.g., sabbatical, conflicts, not sufficiently familiar with topic for purposes of the Prelim, personal reasons); in such cases another examiner must be located or a different topic should be pursued. It is important to ask potential committee members well in advance if they will be able to administer the Prelim. It is recommended that students begin planning their Prelim one semester in advance.

To avoid possible biases and conflicts of interest, it is strongly recommended, but not required, that the student's major research adviser not serve on the Prelim Committee. This may be challenging to avoid if only a small number of faculty are available to administer the Prelim in a given period. In consultation with the HGA, alternate prelim committee members should be considered in such cases before adding a research advisor to the committee. A research advisor is not allowed to chair a student's Prelim committee under any circumstance.

Evaluation, Outcomes, Retakes

Each exam is graded separately. The possible outcomes are:

- **Pass.** Passing the Prelim means getting a pass in all three topical exams. A pass is equivalent to the level of performance meriting a "B" grade on any of the graduate core courses.
- **Contingent pass.** A contingent pass is a type of full pass in which a student passes all portions of the exam, but the committee feels the student would benefit from an additional activity (e.g., teaching 7A for a semester). The committee then gives a pass that is conditional on the contingency (e.g., teaching) being satisfied.
- **Partial Pass.** Partial passes occur when a student passes one or two of the exams only. A retake of the topics not passed is required.
- **Fail.** A fail is when none of the individual topic exams receive a pass. A retake of all topics is required.

If a student does not receive a full pass on Prelim the first time, one additional attempt will be allowed. The 2nd exam will either be on the topic(s) not passed in the case of a partial pass, or on all exams in the event of a fail. The 2nd prelim exam should be taken a minimum of 2 months and a maximum of 6 months after the first attempt. If a student does not receive a full pass on the 2nd attempt, the chair of the exam committee and the HGA will discuss future options with the student. The department will recommend that the student be placed on academic probation if the Prelim is not passed on the 2nd attempt. A student must pass the Prelim within six months after the end of the second academic year. The presence of three faculty members on the committee is designed to ensure exam fairness. Faculty are obligated to report any misgivings about the fairness of the exam to the HGA (or Department Chair, Vice-chair, or Equity Advisor, in the event of a conflict).

Questions

Faculty will make a good faith effort to record questions during each Prelim. Following each exam, faculty will make a good faith effort to send questions transmitted to the GSAO for preserving the record. Note that for various reasons this will result in an exhaustive, but not necessarily complete list of questions. Questions will be available upon request once an entire cohort (e.g., all 2nd year students) have completed their Prelims (including retakes). Students are not allowed to informally share questions that are under embargo; this would be considered a violation of the University student code of conduct.

Official Prelim Advice

- Begin Prelim planning well in advance. It is recommended that planning begin 1 semester prior to the proposed exam date.
- Schedule a Prelim exam time 2 months (or more) in advance. It can be hard to find a day and time that all committee members can make.
- Set aside plenty of time to study without significant distractions. On the flip side, exam preparation can be tedious and may lead to burnout. Students may wish to alternate Prelim studying with research activities so as to mitigate burnout and not feel too disconnected from research.
- Choose topics that are reasonably comfortable. They don't necessarily have to be central to an intended area of research (e.g., a student may enjoy and do well in their stars class, even if their intended area of research isn't related to stars).
- Changing a topic may be necessary if an examiner cannot be found. Planning well in advance can help mitigate such concerns.
- Prior to the Prelim, it is advisable, though not necessary, to meet one-on-one with faculty on the Prelim committee to ask questions and/or clarify their expectations for passing.
- Ask fellow graduate students to administer practice Prelim exams.
- Use Prelim questions from previous years as guidance. Note that while these questions can serve as guidance, there is no guarantee that actual exam questions will resemble previous years' questions. Question construction and style can vary year to year, or within the same year, even for the same faculty member. Creating new, not seen-before questions gets increasingly challenging for faculty as more questions are made available for study and can lead to more creative questions.

PRELIM ADVICE

Past Prelims

[Click here for prelim questions until 2015.](#)

[Click here for prelim questions after 2015.](#)

Advice

[Click here for advice on how to pass your prelim without going nuts.](#)

Preparing for the Prelim

The prelim is an oral exam generally taken at some point during the second year (can be taken as late as December of your 3rd year). It's roughly a "master's exam," in that after passing it (and completing the required coursework), you can pick up your MA. (You can even walk in graduation if you like!) The exam covers three astronomy graduate courses of the student's choosing; each subject is given by a professor of the student's choosing and is 40 minutes long, for a 2-hour-long exam.

Choosing Topics

The prelim consists of three subjects; these almost always correspond to astronomy graduate courses. Most courses can be used as prelim topics, but there are several exceptions that you should know about. In case of doubt, talk to the Head Graduate Adviser (who, if also in doubt, consults the faculty as a whole). You are, in any case, required to complete a Google Form [<https://forms.gle/29kuKcsWox3cxp2x9>] to decide your topics and obtain the Head Graduate Adviser's approval of your topics.

Restrictions include:

- You can't take a prelim in both Radiative Processes (201) and Fluid Dynamics (202), on the grounds that not enough astronomical phenomenology is covered.
- You can't take a prelim in Numerical Techniques (204) (ditto above)
- Prelims in Computational Methods (255) and Plasma Astrophysics (267) are discouraged (ditto)
- You can't take a prelim in both ISM (216) and Star Formation (Steve Stahler's 250) (too much overlap)
- Most seminars (250 classes) can't be used for prelims
- Physics courses generally can't be used for prelims, with possible exceptions
- Starting in 2019, the undergrad course Astro C161 is sufficient background for the cosmology prelim. Physics 229 may also be taken for the cosmology prelim, although it is more advanced and focuses more on research.

Prelim topics do not need to be topics that you plan to concentrate on in your thesis. Some people use the prelim as opportunity to become fluent in their research area, but others use it as a chance to study a topic that they would never otherwise learn in great detail. The choice is yours.

Choosing a Committee

Each professor on the committee is responsible for a single topic. In general, people ask the professor who taught them a subject to administer the prelim for that subject, but this is not required. Any professor can be asked to give any topic, although for a given topic, it makes sense to stick to professors who have taught that course at some point or who work in that area. (There is a UC rule that your research advisor cannot be on your committee; in the Astronomy department, the faculty may strongly encourage you to avoid your research advisor when choosing your committee, out of concerns of fairness, but they will not enforce this rule.)

When choosing a committee, you also have to ask one of the members to be chair. The responsibilities of the chair are:

- To take care of general administration (this includes picking up your case folder before the exam, and dropping it off afterward)
- To make sure that questions are well-posed and fair
- To enforce time limits (40 minutes per topic)
- To establish consensus among committee members about the outcome of the exam

While the chair does have the above duties, in practice they acts just like the other committee members and have no additional authority.

Organizing the Exam Day

You are responsible for coordinating all the logistics of your exam, which includes asking people to serve on your committee, finding a date and time that works for everyone, and reserving a room. Because professors schedules tend to be rather full, it's worthwhile to begin asking people to serve on your committee 2-3 months in advance and to set up a date as soon as you've chosen your committee. If you plan on taking your prelim during the summer, you should start even earlier than this, since some people go away for large parts of the summer.

Once you've arrived for a date and time, you need to book a room. Most prelims occur in Room 419, Room 501B, or Room 219. You should book the room for at least an extra half hour after your exam, to give the committee time to talk with you afterwards and to give you time to clean up.

You also need to let the department administrators know your prelim date, so that they can prepare the appropriate paperwork. They will put an assessment sheet inside your file, for your committee to sign after the exam. After the exam, they will send you a paper that needs to be filed with the university in order to be issued the masters degree.

The only paperwork you need to complete before the exam is the prelim application form which can be found here (Google Form) [<https://forms.gle/29kuKcsWox3cpx2x9>].

Study Materials

Most people study from some combination of the following: class notes, flash cards, and previous prelim questions. Class notes are especially useful if the professor administering a certain topic is the same one who taught you that topic. If not, it is also useful to borrow class notes from someone who did take the course with that professor. Flash cards can be helpful as well, since during the prelim, you'll be expected to have most constants and equations at your fingertips. You should definitely look at previous prelim questions while studying. The administrators have two binders of these in their office (going back to the 1970s!). These can be borrowed essentially indefinitely. A wiki version is under construction. Studying generally occupies the better part of the month prior to the exam. Click here for tips and advice from specific students.

Practicing

Your fellow grad students are happy to help you prepare for the prelim! Most people take a practice prelim, with a fake committee composed of post-prelim grad students; this helps prepare for both the material and the format of the prelim. The practice prelim is generally taken a few days to a week in advance of the actual prelim. It is identical to the prelim except that the committee is composed of grad students and the results don't count.

The practice prelim can be organized only a week or two in advance, but make sure to leave enough time for your fake committee members to make a good list of questions. Your fake committee should be as close to your actual committee as possible; for each professor and topic, you should try to get a grad student who works in that area (with that professor, if possible) and who has taken that course (as a prelim subject, if possible). Most people find the comments of the fake committee extremely useful in preparing for the actual exam.

Your fake committee may include students whose prelim questions are still under the three-month embargo, provided that they don't ask any embargoed questions during your practice prelim.

Don't forget to book a room. And to bring food for your fake committee.

The Day Before

The day prior to your exam there are three things you should do: remind your committee, buy food, and ask Amber to put your folder together. Your committee members will likely not remember the exact date, time, and location of your prelim, so you can avoid a little confusion by sending them a subtle reminder email. It's also nice to ask if they have any specific food preferences. (You can also ask your fake committee for professor-specific food recommendations.) Don't forget to get/make food for your committee; they expect to be fed! Most people bring snack food, healthy and otherwise, and some juice. Also email Amber to let her know that you're taking your prelim so she can put your folder for the chair.

During the Exam

The Exam Process

You should arrive for your prelim a few minutes early, to test that all the white-board markers work and to set out the food. With a little luck, your committee will all arrive on time, and the chair will bring your file. Your file contains your transcript and any other official documents relevant to your progress as a grad student thus far. The committee will then send you out of the room for a few minutes to look over your file and discuss you. They'll call you back in, and then the fun starts!

You get to choose the order of your prelim topics. Conventional grad student wisdom is to put your second best subject first, your worst subject second, and your best subject third. You will tell your committee of the exam order right before the exam starts.

Each professor has 40 minutes to examine you on their subject. Occasionally, the other members will interject with additional questions. You'll be expected to give some qualitative answers and also to do some math and/or plotting on the board. While each professor questions you, one of the other committee members will be recording their questions. In general, nobody makes notes on your performance during the exam.

At the end of the prelim, the committee will again send you out of the room while they decide whether or not you pass. If you let your fellow grads know the time of your prelim, they will show up to cheer you on while you wait. The committee sometimes discusses for only a few minutes, sometimes for much longer. The length of time they talk has very little to do with your competence or performance.

The committee will then call you back in, to tell you whether you passed and their general thoughts on your performance in the exam and as a grad student in general. You can pass or fail all three subjects independently. If you pass all three, they will all sign the assessment paper in

your file and return the file to Dexter. If you fail any or all of the three subjects, you can retake the exam in those subjects; you're expected to wait about a month before retaking the exam. Failing happens occasionally, but in current grad student memory, everyone who's failed the prelim the first time has passed the second time. Once your committee has finished chatting with you, you clean up the food and the boards, and then you're done!

A Little Prelim Wisdom

Everyone has their own prelim wisdom, and it's helpful to ask around to get those gems of knowledge. Here's a little list of things that others have found useful:

- Talk first, write later. If possible, give a concise, conceptual answer to each question before you turn around and start writing on the board. Professors care much more about your conceptual understanding of the topics than they do about your facility with math. Frequently they'll be satisfied with a two-sentence oral answer and will ask you to go on without writing anything down at all!
- However, you shouldn't be afraid of writing on the board, and don't be afraid of taking time to work through calculations. When you do write on the board, write neatly, starting on the top left side of the board, and keep your writing organized. This will help you to successfully identify the inevitable algebra mistakes that you will make.
- A sneaky way to buy some time while thinking about a question is to wait to erase the board until *after* a question is asked (as opposed to erasing it between questions).
- Avoid the statement "I don't know". Instead, when stumped, talk through your thought process; detail what you do know and what you're stuck on. This gives your committee a better idea of your understanding and usually opens a window for further questions that lead you to the answer.
- Don't confuse the above advice with a license to BS. If you have no idea what you're talking about you should not talk for the sake of talking or avoiding saying "I don't know". The point is to show them what you DO know, even if it's not precisely what they asked, so that they can nudge you in the right direction.
- Make sure you understand the question. Ask for clarification if you're not sure. It's very easy to misinterpret a basic question as something fantastically arcane and then go babbling on about something unrelated to avoid saying "I don't know".
- Upon answering a question correctly **do not** proceed to volunteer additional information to make yourself look smart. This will inevitably lead to trickier questions and more detailed nitpicking. Plus, you can very easily be *wrong*!
- Bring a water bottle. Talking for two hours makes you thirsty.

Testimonials

[Click here](#) for personalized prelim wisdom from grad students who have made it through the fiery hoop.

Post-Prelim

Once you passed your prelim, there are just a few things that you need to do. The first is compile the questions you were asked. Your committee will give you their hand-written notes from your exam; these should be typed up and given to Dexter to add to the binders of prelim questions. You should also add them to the wiki once the 3 month embargo is up.

Dexter will also send you a form that should be filed with the university (via Dexter); this form lets the university know when you expect to earn your masters degree. Degrees are issued in both December and May. The form needs to be filed at the beginning of the semester in which you plan to earn the degree. If, for example, you take the prelim at any point during the spring semester and getting the degree asap is important to you, you should file this form in mid-February. If getting the degree asap is not a priority, just fill out this form once you've passed the prelim and your masters will be available at the end of the following semester. For more information, visit

http://www.grad.berkeley.edu/policies/degree_filing_deadlines.shtml [http://www.grad.berkeley.edu/policies/degree_filing_deadlines.shtml].

Now take a vacation; you deserve it!